

EE341 — Fourier problem set

20 questions with full solutions

Fourier series, transforms, properties, convolution, Parseval, sampling — and their use on modulated signals

Built for a **Fourier-heavy quiz**. §1 is the reference you should be able to write from memory in two minutes. §2–§8 are the questions, roughly in increasing difficulty within each group. Solutions follow each question immediately — cover them with a sheet of paper.

Convention: f in Hz throughout, $X(f) = \int x(t)e^{-j2\pi ft} dt$, and $\text{sinc}(u) = \sin(\pi u)/(\pi u)$.

1 The reference sheet

Properties

$ax_1 + bx_2$	$aX_1 + bX_2$
$x(t - t_0)$	$X(f)e^{-j2\pi ft_0}$
$x(t)e^{j2\pi f_0 t}$	$X(f - f_0)$
$x(t)\cos(2\pi f_0 t)$	$\frac{1}{2}X(f - f_0) + \frac{1}{2}X(f + f_0)$
$x(at)$	$\frac{1}{ a }X(f/a)$
$X(t)$	$x(-f)$ (duality)
$x^*(t)$	$X^*(-f)$
x real	$X(-f) = X^*(f)$
$x_1 * x_2$	$X_1 X_2$
$x_1 x_2$	$X_1 * X_2$
dx/dt	$j2\pi f X(f)$
$t x(t)$	$\frac{-1}{j2\pi} \frac{dX}{df}$
$\int_{-\infty}^t x$	$\frac{X(f)}{j2\pi f} + \frac{1}{2}X(0)\delta(f)$

Pairs

$\delta(t)$	1
1	$\delta(f)$
$\cos 2\pi f_0 t$	$\frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$
$\text{rect}(t/T)$	$T \text{sinc}(fT)$
$\text{tri}(t/T)$	$T \text{sinc}^2(fT)$
$2W \text{sinc}(2Wt)$	$\text{rect}(f/2W)$
$e^{-at}u(t)$	$\frac{1}{a + j2\pi f}$
$e^{-a t }$	$\frac{2a}{a^2 + (2\pi f)^2}$
$u(t)$	$\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$
$\text{sgn}(t)$	$\frac{1}{j\pi f}$
$\frac{1}{\pi t}$	$-j \text{sgn}(f)$
$\sum_n \delta(t - nT_s)$	$f_s \sum_k \delta(f - kf_s)$

Parseval $\int |x|^2 dt = \int |X|^2 df$

Areas $X(0) = \int x dt, \quad x(0) = \int X df$

2 Transforms from the definition

Q1. Find the Fourier transform of $x(t) = e^{-at}u(t)$, $a > 0$. Sketch $|X(f)|$, and find the 3 dB bandwidth.

Solution.

$$X(f) = \int_0^\infty e^{-at} e^{-j2\pi ft} dt = \int_0^\infty e^{-(a+j2\pi f)t} dt = \left[\frac{e^{-(a+j2\pi f)t}}{-(a+j2\pi f)} \right]_0^\infty = \frac{1}{a + j2\pi f}.$$

The upper limit vanishes because $|e^{-(a+j2\pi f)t}| = e^{-at} \rightarrow 0$ (this is where $a > 0$ is used).

$$|X(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}}, \quad \angle X(f) = -\arctan \frac{2\pi f}{a}.$$

A Lorentzian peaking at $|X(0)| = 1/a$ and decaying like $1/f$. Half-power ($|X|^2$ down by 2) needs $a^2 + (2\pi f)^2 = 2a^2$, i.e. $2\pi f = a$:

$$B_{3\text{dB}} = \frac{a}{2\pi} \text{ Hz}$$

Note the reciprocity: large a (fast decay in time) $\Rightarrow$ large bandwidth.

Q2. Find the transform of $x(t) = e^{-a|t|}$, $a > 0$, and verify it is real and even. Explain why that was predictable.

Solution. Split the integral:

$$X(f) = \int_{-\infty}^0 e^{at} e^{-j2\pi ft} dt + \int_0^{\infty} e^{-at} e^{-j2\pi ft} dt = \frac{1}{a - j2\pi f} + \frac{1}{a + j2\pi f} = \frac{2a}{a^2 + (2\pi f)^2}.$$

Real and even. **Predictable** because x is real *and* even, and real-even $\Rightarrow$ real-even (the sin part of the kernel integrates against an even function to zero).

Q3. Using $\text{rect}(t) * \text{rect}(t) = \text{tri}(t)$, find the transform of $\text{tri}(t/T)$ without integrating.

Solution. Scale first: $\text{rect}(t/T) * \text{rect}(t/T)$ has width $2T$ and peak equal to the maximum overlap area $= T$, so

$$\text{rect}(t/T) * \text{rect}(t/T) = T \text{tri}(t/T).$$

Transform both sides using $x_1 * x_2 \xrightarrow{\mathcal{F}} X_1 X_2$ and $\text{rect}(t/T) \xrightarrow{\mathcal{F}} T \text{sinc}(fT)$:

$$T \cdot \mathcal{F}\{\text{tri}(t/T)\} = [T \text{sinc}(fT)]^2 \implies \boxed{\text{tri}(t/T) \xrightarrow{\mathcal{F}} T \text{sinc}^2(fT)}$$

Check: $X(0) = T$ should equal $\int \text{tri}(t/T) dt = T$ (triangle of base $2T$, height 1). $\checkmark$

3 Properties

Q4. $x(t)$ has transform $X(f)$. Find the transform of $y(t) = x(2t - 3)$.

Solution. **Factor the argument first** — this is the whole question:

$$y(t) = x\left(2\left(t - \frac{3}{2}\right)\right) : \text{scale by 2, then delay by } \frac{3}{2}.$$

Scaling: $x(2t) \xrightarrow{\mathcal{F}} \frac{1}{2} X(f/2)$. Delaying that by $\frac{3}{2}$:

$$\boxed{Y(f) = \frac{1}{2} X(f/2) e^{-j3\pi f}}$$

Trap. Reading it as “delay by 3 then scale” gives $\frac{1}{2} X(f/2) e^{-j6\pi f}$ — wrong. Always write $x(a(t - t_0))$ explicitly before applying anything.

Q5. Given $\text{rect}(t) \xrightarrow{\mathcal{F}} \text{sinc}(f)$, use duality to find the transform of $\text{sinc}(2Wt)$.

Solution. Duality: $X(t) \xrightarrow{\mathcal{F}} x(-f)$. With $x = \text{rect}$, $X = \text{sinc}$:

$$\text{sinc}(t) \xrightarrow{\mathcal{F}} \text{rect}(-f) = \text{rect}(f) \quad (\text{rect even}).$$

Now scale: $\text{sinc}(2Wt)$ is $\text{sinc}(at)$ with $a = 2W$, so

$$\boxed{\text{sinc}(2Wt) \xrightarrow{\mathcal{F}} \frac{1}{2W} \text{rect}\left(\frac{f}{2W}\right)}$$

i.e. an ideal lowpass of width $2W$ and height $1/2W$. **Check:** $X(0) = 1/2W$ must equal $\int \text{sinc}(2Wt) dt = 1/2W$. ✓

Q6. Find the transform of $t e^{-at} u(t)$ two ways: (a) by the frequency-differentiation property, (b) by recognising a convolution.

Solution. (a) $t x(t) \xleftrightarrow{\mathcal{F}} -\frac{1}{j2\pi} \frac{dX}{df}$. With $X = (a + j2\pi f)^{-1}$,

$$\frac{dX}{df} = -\frac{j2\pi}{(a + j2\pi f)^2} \implies \mathcal{F}\{t e^{-at} u(t)\} = \frac{-1}{j2\pi} \cdot \frac{-j2\pi}{(a + j2\pi f)^2} = \boxed{\frac{1}{(a + j2\pi f)^2}}$$

(b) $(e^{-at} u) * (e^{-at} u)(t) = \int_0^t e^{-a\tau} e^{-a(t-\tau)} d\tau = t e^{-at}$ for $t > 0$. So the signal is that self-convolution, whose transform is $X^2 = (a + j2\pi f)^{-2}$. ✓ Same answer, and (b) is faster if you spot it.

Q7. $x(t)$ is real with $X(f)$ known. Express the transforms of the even and odd parts of x in terms of X .

Solution. $x_e = \frac{1}{2}[x(t) + x(-t)]$ and $x_o = \frac{1}{2}[x(t) - x(-t)]$. Since $x(-t) \xleftrightarrow{\mathcal{F}} X(-f) = X^*(f)$ for real x :

$$x_e \xleftrightarrow{\mathcal{F}} \frac{1}{2}[X(f) + X^*(f)] = \text{Re}\{X(f)\}, \quad x_o \xleftrightarrow{\mathcal{F}} \frac{1}{2}[X(f) - X^*(f)] = j \text{Im}\{X(f)\}.$$

So for a real signal: **the even part carries the real part of the spectrum, the odd part carries the imaginary part.** That is why real-even $\Rightarrow$ real spectrum and real-odd $\Rightarrow$ imaginary spectrum.

4 Fourier series

Q8. A periodic rectangular pulse train has amplitude A , pulse width τ , period T_0 , centred on $t = 0$. Find c_n , sketch the line spectrum, and say where the envelope nulls fall.

Solution.

$$c_n = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} A e^{-j2\pi n f_0 t} dt = \frac{A}{T_0} \cdot \frac{e^{j\pi n f_0 \tau} - e^{-j\pi n f_0 \tau}}{j2\pi n f_0} = \frac{A}{T_0} \cdot \frac{\sin(\pi n f_0 \tau)}{\pi n f_0}$$

$$\boxed{c_n = \frac{A\tau}{T_0} \text{sinc}\left(\frac{n\tau}{T_0}\right)}$$

Lines at $f = n f_0 = n/T_0$, spacing $1/T_0$; the envelope is a sinc of first null at $f = 1/\tau$. Nulls therefore fall on lines $n = T_0/\tau, 2T_0/\tau, \dots$ (whenever $n\tau/T_0$ is a nonzero integer). $c_0 = A\tau/T_0$ is the duty cycle times A , i.e. the average — a free check.

Two knobs, two effects. T_0 sets the *line spacing* ($1/T_0$); τ sets the *envelope width* ($1/\tau$). Narrower pulses $\Rightarrow$ wider envelope; longer period $\Rightarrow$ denser lines. Letting $T_0 \rightarrow \infty$ with τ fixed makes the lines merge into the continuous $\tau \text{sinc}(f\tau)$ — that is the series becoming the transform.

Q9. Derive the Fourier transform of the impulse train $s(t) = \sum_n \delta(t - nT_s)$ from its Fourier series.

Solution. s is periodic with period T_s , so $s(t) = \sum_n c_n e^{j2\pi n f_s t}$ with $f_s = 1/T_s$ and

$$c_n = \frac{1}{T_s} \int_{-T_s/2}^{T_s/2} \delta(t) e^{-j2\pi n f_s t} dt = \frac{1}{T_s} = f_s$$

(only the $n = 0$ impulse lies in the window; the sifting property gives $e^0 = 1$). So all coefficients are equal:

$$s(t) = f_s \sum_n e^{j2\pi n f_s t} \xleftrightarrow{\mathcal{F}} \boxed{S(f) = f_s \sum_k \delta(f - k f_s)}$$

using $e^{j2\pi n f_s t} \xleftrightarrow{\mathcal{F}} \delta(f - n f_s)$. **An impulse train transforms to an impulse train**, with *reciprocal* spacing and height f_s .

Q10. Find the Fourier series of the ± 1 square wave $c(t)$ of period T_c , taking $t = 0$ at the centre of a $+1$ interval. Which symmetry kills the even harmonics, and which makes the answer come out in cosines?

Solution.

$$c_n = \frac{1}{T_c} \left[\int_{-T_c/4}^{T_c/4} e^{-j2\pi n f_c t} dt - \int_{T_c/4}^{3T_c/4} e^{-j2\pi n f_c t} dt \right] = \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right),$$

which is 0 for even n and $\pm 2/n\pi$ for odd n . Pairing $\pm n$:

$$c(t) = \frac{4}{\pi} \left[\cos 2\pi f_c t - \frac{1}{3} \cos 6\pi f_c t + \frac{1}{5} \cos 10\pi f_c t - \dots \right].$$

Half-wave symmetry $c(t+T_c/2) = -c(t)$ kills the even harmonics (origin-independent). **Evenness** about the chosen origin makes it cosines only. Two different symmetries, two different jobs.

5 Convolution and multiplication

Q11. Find the transform of $y(t) = x(t) \cos^2(2\pi f_c t)$ in terms of $X(f)$, and sketch it for X a triangle of half-width W .

Solution. Do not convolve — use the identity first:

$$\cos^2 2\pi f_c t = \frac{1}{2} + \frac{1}{2} \cos(4\pi f_c t) \implies y = \frac{1}{2}x(t) + \frac{1}{2}x(t) \cos(4\pi f_c t).$$

$$Y(f) = \frac{1}{2}X(f) + \frac{1}{4}X(f - 2f_c) + \frac{1}{4}X(f + 2f_c)$$

Sketch: a triangle of half-height at DC, and two quarter-height triangles at $\pm 2f_c$. Note there is **nothing** at $\pm f_c$. Squaring the carrier moved everything to DC and $2f_c$ — which is exactly the mechanism in coherent detection.

Q12. x_1 is bandlimited to W_1 and x_2 to W_2 . What is the bandwidth of $x_1(t)x_2(t)$? Prove it.

Solution. Multiplication in time is convolution in frequency: $Y = X_1 * X_2$. The support of a convolution is the sum of the supports (widths add), so with X_1 on $[-W_1, W_1]$ and X_2 on $[-W_2, W_2]$,

$$Y(f) = 0 \text{ for } |f| > W_1 + W_2 \implies B = W_1 + W_2$$

Corollary used constantly: $x^2(t)$ has bandwidth $2W$ — which is why the square-law modulator's baseband group is $2W$ wide, and why $f_c > 3W$ is needed there.

Q13. An ideal lowpass filter of cutoff W and unit gain is fed $x(t) = \delta(t)$. Find the output, and explain why the filter is not realisable.

Solution. $Y(f) = H(f) \cdot 1 = \text{rect}(f/2W)$, so $y(t) = h(t) = 2W \text{sinc}(2Wt)$. **Not realisable** for two reasons: (i) $h(t) \neq 0$ for $t < 0$ — the filter responds *before* the impulse arrives, violating causality; (ii) $\int |h| dt = \infty$ (the sinc decays only as $1/t$), so it is not BIBO stable. Real filters trade a finite transition band for causality.

6 Parseval and energy

Q14. Compute $\int_{-\infty}^{\infty} \text{sinc}^2(2Wt) dt$ without doing a hard integral.

Solution. From Q5, $\text{sinc}(2Wt) \xleftrightarrow{\mathcal{F}} \frac{1}{2W} \text{rect}(f/2W)$. Parseval:

$$\int \text{sinc}^2(2Wt) dt = \int \left| \frac{1}{2W} \text{rect}\left(\frac{f}{2W}\right) \right|^2 df = \frac{1}{4W^2} \cdot 2W = \boxed{\frac{1}{2W}}$$

The frequency-domain integral is the area of a rectangle. That is the whole point of Parseval: do the integral in whichever domain the function is simpler.

Q15. For $x(t) = e^{-at}u(t)$, what fraction of the total energy lies within the 3 dB bandwidth $|f| \leq a/2\pi$?

Solution. Total energy, easiest in time: $E = \int_0^\infty e^{-2at} dt = \frac{1}{2a}$.

In-band energy, in frequency with $|X|^2 = (a^2 + (2\pi f)^2)^{-1}$. Substitute $u = 2\pi f/a$, so $df = \frac{a}{2\pi} du$ and the limits become ∓ 1 :

$$E_{\text{in}} = \int_{-a/2\pi}^{a/2\pi} \frac{df}{a^2 + (2\pi f)^2} = \frac{1}{2\pi a} \int_{-1}^1 \frac{du}{1 + u^2} = \frac{1}{2\pi a} \left[\arctan u \right]_{-1}^1 = \frac{1}{2\pi a} \cdot \frac{\pi}{2} = \frac{1}{4a}.$$

$$\text{fraction} = \frac{1/4a}{1/2a} = \boxed{50\%}$$

Worth remembering: for a one-sided exponential, the 3 dB bandwidth contains exactly half the energy. (Sanity check: letting the limits $\rightarrow \infty$ gives $\frac{1}{2\pi a} \cdot \pi = \frac{1}{2a} = E$ ✓.)

7 Sampling

Q16. $x(t) = 2 + 3\cos(2\pi \cdot 1000t) + \cos(2\pi \cdot 3000t)$ is sampled at $f_s = 4$ kHz and reconstructed with an ideal LPF of cutoff 2 kHz.

- Is Nyquist satisfied?
- What comes out?

Solution. (a) Highest frequency is 3 kHz, so Nyquist needs $f_s > 6$ kHz. With $f_s = 4$ kHz, **aliasing occurs**.

(b) Sampling replicates every line at $\pm f + kf_s$. Track each component into $[-2, 2]$ kHz:

component	replicas	lands in band at
DC (2)	$0, \pm 4, \dots$	0
1 kHz	$1, 3, -3, 5, \dots$	1 kHz
3 kHz	$3, 3 - 4 = -1, -3 + 4 = 1, \dots$	1 kHz (aliased!)

The 3 kHz tone folds down onto 1 kHz and *adds* to the genuine one. Check on samples: at $t = n/4000$, $\cos(2\pi \cdot 3000 n/4000) = \cos(1.5\pi n) = \cos(0.5\pi n) = \cos(2\pi \cdot 1000 n/4000)$ — identical sample sequences, so they are indistinguishable.

$$\boxed{y(t) = 2 + 4\cos(2\pi \cdot 1000 t)}$$

The 3 kHz information is gone irretrievably, and the 1 kHz amplitude is corrupted from 3 to 4. An anti-aliasing filter before sampling would have removed the 3 kHz tone and left $2 + 3\cos$.

Q17. Prove $X_s(f) = f_s \sum_k X(f - kf_s)$ for impulse sampling, and state the reconstruction filter needed.

Solution. $x_s(t) = x(t) \sum_n \delta(t - nT_s)$. Multiplication in time $\Rightarrow$ convolution in frequency, and by Q10 the impulse train has transform $f_s \sum_k \delta(f - kf_s)$:

$$X_s(f) = X(f) * f_s \sum_k \delta(f - kf_s) = f_s \sum_k X(f - kf_s),$$

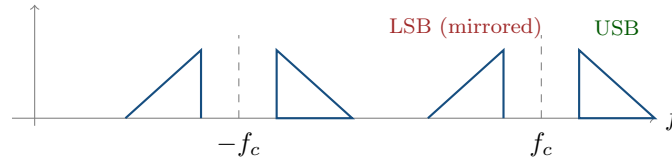
using $X(f) * \delta(f - f_0) = X(f - f_0)$. **Reconstruction:** an ideal LPF with cutoff between W and $f_s - W$ and gain $T_s = 1/f_s$ (to undo the f_s scaling), which yields $x(t)$ exactly when $f_s > 2W$.

8 Fourier on modulated signals

Q18. A message has the *asymmetric* spectrum $X(f)$ shown: zero except on $0 < f < W$ where it rises linearly (plus its Hermitian mirror on $-W < f < 0$). Sketch the spectrum of $x(t) \cos 2\pi f_c t$ and mark clearly which sideband is inverted.

Solution. $X_c(f) = \frac{1}{2}X(f - f_c) + \frac{1}{2}X(f + f_c)$: two half-height copies of the *whole* X , one shifted to $+f_c$ and one to $-f_c$.

Around $+f_c$: the part of X that lived on $0 < f < W$ now sits on $f_c < f < f_c + W$ — the **upper sideband**, un-inverted. The part that lived on $-W < f < 0$ now sits on $f_c - W < f < f_c$ — the **lower sideband**, and because it came from the negative-frequency half it is the **mirror image**.



Drawing symmetric triangles here loses marks — an asymmetric X is chosen precisely to test whether you know the lower sideband is reversed.

Q19. Show that $\mathcal{F}\{x(t) + j\hat{x}(t)\} = 2X(f)u(f)$, and use it to explain in one line why SSB works.

Solution. $\hat{x} \xleftrightarrow{\mathcal{F}} -j \operatorname{sgn}(f)X(f)$, so

$$\mathcal{F}\{x + j\hat{x}\} = X(f) + j(-j \operatorname{sgn}(f)X(f)) = X(f)(1 + \operatorname{sgn} f) = 2X(f)u(f),$$

since $1 + \operatorname{sgn} f$ is 2 for $f > 0$ and 0 for $f < 0$.

Why SSB works: $x + j\hat{x}$ contains only the message's positive frequencies, so modulating it up to f_c and taking the real part deposits content only *above* f_c (and its mirror below $-f_c$) — one sideband, half the bandwidth, no information lost because Hermitian symmetry made the other sideband redundant anyway.

Q20. A channel has $H(f) = Ke^{-j2\pi f t_d}$ over the signal band. Show the output is undistorted. What goes wrong if instead $\angle H(f) = -2\pi f t_d + \theta_0$ with constant $\theta_0 \neq 0$, for a *bandpass* signal?

Solution. $Y(f) = Ke^{-j2\pi f t_d}X(f) \Rightarrow y(t) = Kx(t - t_d)$ by the time-shift property: a scaled, delayed copy — **distortionless** by definition.

With an extra constant θ_0 : for a bandpass signal $x_{bp} = \operatorname{Re}\{x_\ell e^{j2\pi f_c t}\}$, the extra phase multiplies the complex envelope by $e^{j\theta_0}$, giving

$$y(t) = K \operatorname{Re}\{x_\ell(t - t_d)e^{j\theta_0}e^{j2\pi f_c(t - t_d)}\}.$$

The **envelope** $|x_\ell|$ is **unchanged**, so an envelope detector sees nothing wrong. But a *coherent* detector does: the recovered signal is scaled by $\cos \theta_0$, and vanishes at $\theta_0 = 90^\circ$. Constant phase offset is harmless to AM envelope detection and fatal to coherent detection — a favourite comparison question.

Q21. Find and sketch the spectrum of $y(t) = x(t)s(t)$ where $s(t)$ is the unipolar switching function $s(t) = \frac{1}{2} + \frac{2}{\pi} \cos 2\pi f_c t - \frac{2}{3\pi} \cos 6\pi f_c t + \dots$ and x is bandlimited to W . Which term does a BPF at f_c keep, and what modulation is it?

Solution. Multiply term by term and use the modulation property on each:

$$Y(f) = \underbrace{\frac{1}{2}X(f)}_{\text{baseband}} + \underbrace{\frac{1}{\pi}[X(f - f_c) + X(f + f_c)]}_{\pm f_c} - \underbrace{\frac{1}{3\pi}[X(f - 3f_c) + X(f + 3f_c)]}_{\pm 3f_c} + \dots$$

(each $\frac{2}{n\pi} \cos$ contributes $\frac{1}{n\pi}$ per copy). A BPF at f_c of width $2W$ keeps $\frac{2}{\pi}x(t) \cos 2\pi f_c t$: **DSB-SC**.

But the real switching modulator gives DSB-AM, because its input is $x(t) + A \cos 2\pi f_c t$, not $x(t)$ alone. The DC term $\frac{1}{2}$ of $s(t)$ then multiplies the *carrier*, contributing $\frac{A}{2} \cos 2\pi f_c t$ — an impulse pair at $\pm f_c$ that survives the filter. Read the input carefully before deciding SC or AM.

Q22. $x(t)$ is real and bandlimited to W . Show that $x(t) \cos 2\pi f_c t$ and $x(t) \sin 2\pi f_c t$ occupy the same band but are orthogonal over a carrier period, and say why that matters.

Solution. Same band: both transforms are $\frac{1}{2}$ -scaled copies of X at $\pm f_c$ (the sine version carries $\mp \frac{1}{2j}$ factors), so both occupy $[f_c - W, f_c + W]$ and its mirror. Magnitudes are identical; only phases differ by 90° .

Orthogonal: over one carrier period,

$$\frac{1}{T_c} \int_{T_c} \cos 2\pi f_c t \sin 2\pi f_c t \, dt = \frac{1}{2T_c} \int_{T_c} \sin(4\pi f_c t) \, dt = 0.$$

Why it matters: two independent messages can be sent on the same carrier frequency, in the same band, and separated perfectly at the receiver — one recovered by multiplying by $\cos$, the other by $\sin$. That is quadrature multiplexing / QAM, and it is the same orthogonality that makes x_I and x_Q independent channels.

If you only drill six things:

1. Modulation property, with the $\frac{1}{2}$'s, applied to an *asymmetric* X (Q18).
2. $x(at - b)$: factor to $x(a(t - b/a))$ before touching it (Q4).
3. Impulse train $\xleftrightarrow{\mathcal{F}}$ impulse train, hence the sampling replication formula (Q10, Q17).
4. Parseval to dodge a hard integral (Q15).
5. Widths add under convolution; bandwidths add under multiplication (Q13).
6. $x + j\hat{x} \xleftrightarrow{\mathcal{F}} 2Xu(f)$ — one line, and it explains SSB and the analytic signal at once (Q19).